NDANIEL MAHULAE

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Fresh graduate in Computer Engineering from Brawijaya University with strong analytical thinking and a passion for exploring and learning new technologies. I enjoy breaking down complex problems, identifying patterns, and developing practical solutions through a systematic approach. Interested in Artificial Intelligence, particularly Computer Vision, with practical experience in image processing, object detection, OCR, and deep learning through academic and personal projects. Seeking opportunities to apply and further develop my skills in building intelligent vision-based solutions for real-world applications.

Education Level

Brawijaya University Aug 2022 – Jul 2026

Bachelor's Degree in Computer Engineering (GPA 3.70/ 4.00)

Undergraduate Thesis Project :

- Developed a passport Machine-Readable Zone (MRZ) extraction and validation system using Mask R-CNN for MRZ detection and EasyOCR for character recognition, based on ICAO Doc 9303 standards.
- Implemented an end-to-end computer vision pipeline on Raspberry Pi 4, integrating image processing, MRZ parsing, semantic post-processing, and check-digit validation for offline edge deployment.

Working Experience

PT Kilang Pertamina Internasional – VP Information Technology Jul 2025 – Sep 2025

Internship Student at Information Technology Division

- Processed and validated IT requirements and task data from the APPIT (Application IT Requirement) system across 3 divisions (BR, BDM, and BITS) and their progress.
- Analyzed requirement statuses and identified bottlenecks in the approval and initiation stages, with requirements decreasing from 159 to 150 and completed tasks increasing from 3 to 9 during the monitoring period.
- Developed a Looker Studio dashboard to visualize requirement and task progress, supporting cross-divisional monitoring and data transparency. Dashboards

Organizational Experience

PTIIK Organization of Open Source (POROS) Jan 2023 – Feb 2025

Staff of Backend

- Facilitated study group activities and supported peer learning while contributing to the development of the organization's website.
- Supported organizational events as a Public Relations committee member by distributing attendance and registration forms, confirming attendance with 25 of 65 registrants, recording 34 attendees.

Indonesia Creative Cities Festival (ICCF) Nov 2025 – Nov 2025

Staff of Registration Clerk

- Managed visitor registration and attendance records while providing registration information and assistance throughout the event.
- Assisted with event logistics and attendee services to ensure smooth event operations.

Dudukbaca Aug 2025 – Jul 2026

Staff of Social Media

- Managed social media administration across 4 platforms: Instagram, X, TikTok, and Threads, including uploading scheduled content and responding to audience messages.
- Developed 4 content plans/content briefs, covering video and educational content, to support the organization's social media activities.
- Published 34 scheduled posts with posting responsibilities rotating across different social media platforms on a weekly basis.
- Participated in 14 Lapak shifts, documenting the reading space and books through photos for Instagram Stories and also engaging with readers.

Bidan Membantu May 2025 – Present

Staff & PIC of Data

- Cleaned and validated attendance data from 2 programs: Langkah Ibu and Webinar, then developed interactive dashboards using Looker Studio and Streamlit to support monitoring and reporting.
- Analyzed Instagram performance data, identifying a 71.1% increase in reach by July 2026 compared with the initial reporting period, supporting evaluation of digital outreach performance.

Dashboards: Langkah Ibu Dashboard · Webinar Dashboard

Skills and Courses

- **Technical Skills :**
  - Programming: Python
  - Artificial Intelligence: Machine Learning, Deep Learning, NLP
  - Computer Vision: Image Processing, OpenCV, Object Detection, OCR, Mask R-CNN, EasyOCR
  - Frameworks & Deployment: TensorFlow, PyTorch, Docker, MLflow
  - Data & Visualization: Data Analysis, Data Visualization, Tableau, Looker Studio
  - Tools: GitHub
- **Courses :**
  - Belajar Dasar Data Science — Dicoding Indonesia
  - Belajar Machine Learning untuk Pemula — Dicoding Indonesia
  - Belajar Fundamental Deep Learning— Dicoding Indonesia
  - Membangun Proyek Deep Learning Tingkat Mahir — Dicoding Indonesia

Projects

Klasifikasi dan Deteksi Penyakit Daun Jagung | Academic Project, Brawijaya University | Oct 2025 – Dec 2025

- Developed a deep learning-based system to classify and detect common corn leaf diseases, including blight, rust, and leaf spot.
- Applied MobileNetV2 with fine-tuning and data management techniques to develop a lightweight image classification solution.
- Focused on computational efficiency for potential deployment on resource-constrained devices. [GitHub Repository]

Klasifikasi Jenis Makanan | Academic Project | Apr 2025 – May 2025

- Developed an image classification model using DenseNet201 pretrained on ImageNet to classify 12 selected food categories from the Food-101 dataset.
- Applied data augmentation and fine-tuning to improve image classification performance.
- Converted the trained model into TensorFlow Lite, SavedModel, and TensorFlow.js formats for deployment across different environments. [GitHub Repository]

Intensive Camp ML – Car Tracking | SpecialSkill | Jun 2025 – Present

- Developed a video-based vehicle tracking system using object detection and object tracking techniques.
- Implemented bounding boxes and unique IDs to track vehicles across video frames.
- Applied computer vision concepts as part of an intensive machine learning training program. [GitHub Repository]

Machine Learning Operations – Topic Modeling | Academic Project, Brawijaya University | Feb 2025 – Jun 2025

- Built an end-to-end topic modeling pipeline for 10,989 PubMed research articles, from web scraping and text preprocessing to model deployment and monitoring.
- Applied Scrapy, NLTK, BERT embeddings, and BERTopic, generating 173 topics with a coherence score of 0.371.
- Developed and containerized a Flask API using Docker, with experiment tracking through MLflow & DVC and monitoring using Prometheus & Grafana. [Github Repository]

Analisis Sentimen | Personal Project | Sep 2025 – Oct 2025

- Developed an Indonesian sentiment analysis pipeline for YouTube gaming comments using XLM-RoBERTa.
- Performed text preprocessing, including tokenization and slang normalization, to handle noisy, informal, and multilingual comments.
- Evaluated model performance on text containing typos, abbreviations, slang, and informal expressions. [GitHub Repository]

LANGUAGES

- **Bahasa** (Native/ Mother Tongue)
- **English** (Beginner)
- **Japan (Beginner/** N4**)**